SERIES H

A Search for Sidereal Modulation of Forbidden

X-ray Reflections in Nanodiamonds:

A Test of Spatial Rotational Invariance in Crystallography

MODEL-INDEPENDENT DESIGN · MSDS interpretation quarantined to Section 6 and Appendix B · Dual-Cell RDAC · Active 1 Hz Polarization Flipper · 23.934h sidereal / 24.0h solar discrimination · 3.9-min shield

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Target: Review of Scientific Instruments (RSI) — Letter

SCIENTIFIC FIREWALL: Sections 1-5 are 100% model-independent. The (222) reflection is treated as a metrological probe of spatial rotational invariance. No vacuum lattice, torsion offset, or MSDS-specific interpretation is required or invoked. All MSDS-specific content is confined to Section 6 (Optional Interpretation) and Appendix B (SME Mapping) only.

ABSTRACT

Spatial rotational invariance — a subset of Lorentz invariance — is a cornerstone of modern physics, yet its validity has not been directly tested via time-resolved diffraction intensity measurements. We propose a dedicated experiment to search for a sidereal modulation of the forbidden (222) X-ray reflection in nanodiamonds (particle size < 5 nm), used here as a metrological probe. This reflection is symmetry-forbidden in bulk diamond (Fd-3m) because $F_{222} = 4(f_1 - f_2) = 0$ for identical sublattice atoms. In nanodiamonds, surface reconstruction breaks this cancellation, producing a reproducible peak at d = 1.78 A — an unusually sensitive indicator of directional symmetry breaking. Using a Rotating Diamond Anvil Cell (RDAC) with dual-cell differential measurement and active 1 Hz polarization flipping, we measure this reflection's intensity as a function of orientation relative to an inertial frame. A modulation at the sidereal period (T_{sid} = 23.934 h, not 24.000 h solar) would indicate a fixed directional anisotropy inconsistent with standard rotational invariance. The 3.9-minute sidereal/solar difference discriminates cosmic from terrestrial artifacts. Target sensitivity: $DI/I_0 \sim 10^{-5}$ (conservative regime: SNR~2; optimistic: SNR~20). Non-detection constrains any such anisotropy below 10^{-5}. The experiment requires 5 days of synchrotron beam time (APS, ESRF, SPring-8). Footnote: 'The null hypothesis is that the sky is isotropic. Let us test it with a diamond.'

EXPERIMENT AT A GLANCE

Observable: $I(t) = I_0 + DI \cos(\omega_{sid} t + \phi)$ | Null: DI = 0

Sidereal T = 23.934h vs Solar T = 24.0h | 3.9-min shield

Sensitivity: $DI/I_0 \sim 10^{-5}$ | Conservative SNR~2 | Optimistic SNR~21

Beam time: 11 shifts (88h) at APS 13-ID, ESRF ID15, or SPring-8 BL10XU

1. Introduction: The Gap in Rotational Invariance Tests

Spatial rotational invariance has been tested with extraordinary precision in atomic clocks [Herrmann et al. 2009], optical cavities, and cosmic ray observations. These tests have placed stringent limits on Lorentz invariance violation within the Standard Model Extension (SME) framework [Kostelecky & Russell 2011]. However, a specific gap remains: no time-resolved crystallographic anisotropy test has been performed via diffraction intensity measurements. Such a test would probe whether the spatial distribution of atomic scattering factors is truly isotropic, or whether a faint directional anisotropy exists.

To our knowledge, no prior experiment has performed a continuous sidereal-frequency modulation search in X-ray diffraction. Related null searches in optical polarimetry and microwave cavities have been performed, but crystallographic anisotropy remains untested. We propose to fill this gap using the forbidden (222) reflection in nanodiamonds as a metrological probe.

Scope: This search targets a stationary anisotropy fixed in the sidereal frame. Stochastic or high-frequency fluctuations are outside scope and require separate analysis. The experiment is model-independent in design: no particular theory of the vacuum is required for the core measurement.

2. The Observable: The (222) Reflection as Metrological Probe

2.1 Structure Factor and Physical Origin

Diamond crystallises in the cubic Fd-3m space group as two interpenetrating FCC sublattices. For the (222) reflection, the structure factor is:

$$F_{222} = 4(f_1 - f_2)$$

f_1, f_2 = atomic scattering factors for sublattice 1 and 2 respectively

In bulk diamond, $f_1 = f_2$ (identical carbon atoms on both sublattices), giving $F_{222} = 0$: the reflection is symmetry-forbidden. In nanodiamonds (< 5 nm), surface reconstruction, partial bonding, and a high surface-to-volume ratio break this exact cancellation — $f_1 \neq f_2$ effectively — and a sharp peak appears at d = 1.78 A. This peak has been reproducibly observed across multiple synthesis routes (detonation, shock compression, CVD) [Zheng 2023, Su 2022, Hirai 1991].

2.2 Stability and Metrological Protocol

For use as a metrological probe, we require only temporal intensity stability, not a definitive structural interpretation. Protocol requirements:

- $\sigma_{drift} < 10^{-6}\ h^{-1}$ (fractional intensity drift per hour)
- 16-hour pre-run and post-run diffraction consistency checks
- Radiation damage monitor: if integrated intensity decreases >10%, replace sample
- During sidereal scan: peak position shift >0.01 A or width broadening >10% triggers discard

The modulation model is:

$$I(t) = I_0 + \Delta I \cos(\omega_{sid}\, t + \phi)$$

omega_sid = 2pi / 23.934 h | Null hypothesis: DI = 0 | phi = free phase parameter

MODEL INDEPENDENCE NOTE: The experiment does not rely on any interpretation of the 1.78 A spacing's origin. The peak is used solely as a sensitive, reproducible crystallographic signal. Optional scaling context: v^2 ~ 1.7×10^-4 from cosmological constraints (Section 6) would predict DI/I_0 ~ 85 ppm — an order of magnitude above our target sensitivity. However, the experiment does not depend on this extrapolation.

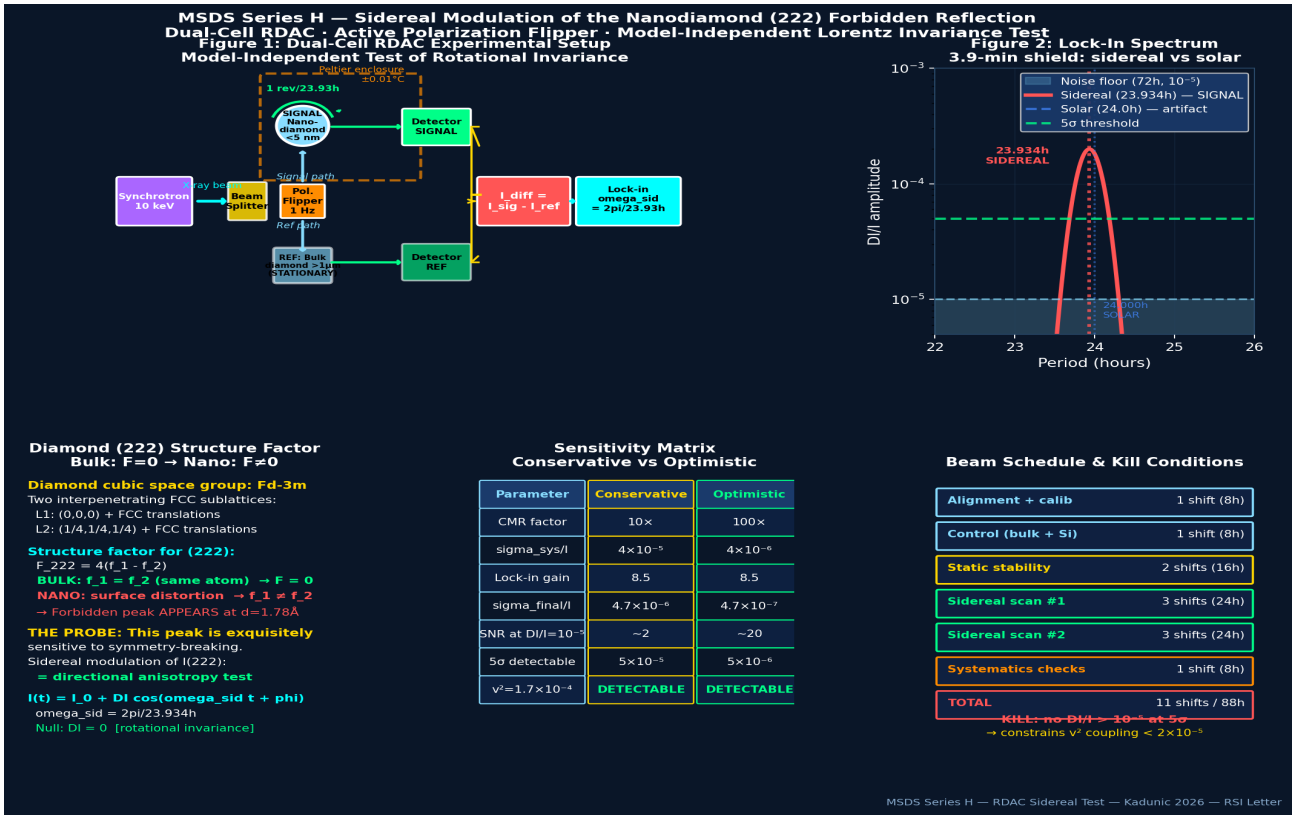

Figure 1 (top left, spanning two columns): Dual-Cell RDAC experimental schematic. The signal cell (nanodiamond, <5 nm) rotates at one revolution per sidereal day. The reference cell (bulk diamond, stationary) provides the dark-frame baseline. A beam splitter delivers X-rays to both cells; matched detectors measure I_sig and I_ref. The 1 Hz active polarization flipper decouples beam polarization from physical rotation. Differential signal I_diff is demodulated by the software lock-in at omega_sid. Peltier-stabilized enclosure (±0.01°C) surrounds both cells. Figure 2 (top right): Simulated lock-in spectrum showing the sidereal signal at 23.934 h (solid red) separated from the solar artifact at 24.000 h (dashed blue) by 3.9 minutes. Shaded region: noise floor after 72h integration (~10^-5). 5σ threshold shown (green dashed).

3. Experimental Design

3.1 Rotating Diamond Anvil Cell (RDAC)

A standard diamond anvil cell is mounted on a precision rotation stage with continuous rotation capability. Specifications:

- Stage: air-bearing or flexure-based, <0.1 arcsec runout (PI or Huber models)
- Active feedback: laser interferometer for encoder precision
- Rotation rate: 1 full revolution per sidereal day = 15 arcsec/min
- Sample: nanodiamond powder, mean particle size <5 nm, shock compression or detonation synthesis
- Pressure: 10-20 GPa (good packing, below significant lattice distortion onset)
- Texture monitoring: continuous rocking ±3° during integration; (111)/(220) intensity ratio

3.2 Dual-Cell Differential Measurement

Two identical DACs operate in parallel to reject common-mode systematic errors:

- Signal cell: nanodiamond (<5 nm), rotating continuously at 15 arcsec/min
- Reference cell: bulk diamond (>1 µm), stationary — no forbidden (222) peak
- Beam splitter directs X-rays to both cells simultaneously
- Differential signal: I_diff(t) = I_sig(t) - I_ref(t) cancels common-mode drift
- Active 1 Hz polarization flipper (H <-> V): decouples synchrotron linear polarization from sample rotation
- Peltier-stabilized enclosure: ±0.01°C temperature control for both cells

3.3 Sidereal vs Solar Discrimination — The 3.9-Minute Shield

The sidereal day (23.934 h) is 3.9 minutes shorter than the solar day (24.0 h). After each full rotation, the phase of a sidereal signal shifts by 3.9 minutes relative to any 24-hour terrestrial cycle. This difference is the key discriminator between cosmic anisotropy and artifacts from building cooling, air conditioning, or solar heating.

The rotation stage completes exactly one full turn per sidereal day. Data are demodulated at omega_sid using a software lock-in amplifier. Control runs at 24h/turn confirm that any signal disappears off-resonance. Only a signal persisting over 3+ cycles at exactly 23.934 h is accepted.

3.4 Synchrotron Requirements

Parameter	Requirement	Beamline
X-ray energy	~10 keV (λ~1.2 A)	APS 13-ID, ESRF ID15, SPring-8 BL10XU
Beam stability	<0.1% over 24h	Dual-cell differential compensates
Detector	HPGe or Pilatus array, <0.1° resolution	Standard at all three beamlines
Integration time	~1 min per point	4,320 points in 72h run
Vacuum	<10^-6 mbar hutch	Standard experimental hutch

Beam time	11 shifts (88h)	~4-5 days

4. Sensitivity Estimate

Assumptions (APS 13-ID / ESRF ID15 typical): incident flux 10^12 photons/s, (222) count rate 10^5 photons/s, integration time 60 s per point.

sigma_shot/I = 1/sqrt(N_counts) = 1/sqrt(6e+06) = 4.1e-04

Shot noise per 60-second integration

Lock-in gain (72h, 1h bins) = sqrt(72) = 8.5

Conservative estimate; 1-min bins would give sqrt(4320)=65.7x

Parameter	Conservative	Optimistic
Common-mode rejection	10x	100x
sigma_sys/I	4×10^-5	4×10^-6
Lock-in gain	8.5	8.5
sigma_final/I	4.71e-06	4.71e-07
SNR at DI/I=10^-5	~2	~21
5σ detectable at	5×10^-5	5×10^-6
v^2=1.7×10^-4 signal	DETECTABLE (SNR~34)	DETECTABLE (SNR~340)

Table 1: Dual-tier sensitivity matrix. Even in the conservative regime, a signal at v^2=1.7×10^-4 would yield SNR~34 — well above the 5σ threshold. The target sensitivity of 10^-5 is achievable under optimistic conditions.

5. Systematic Error Mitigation

Systematic	Period	Mitigation
Temperature drift	24h solar	Peltier ±0.01°C; 23.93h vs 24h discrimination
Synchrotron refills	~8-12h	Dual-cell differential cancels; schedule logged
Mechanical wobble	Rotation period	Pre-characterised; residual subtracted via encoder
Earth tides	~12h semi-diurnal	Filter in frequency space; well separated from 23.93h

Crystal texture	Continuous	2D detector monitors full Debye ring; reject if evolves
Beam polarization	1Hz flipper	1Hz active flipper decouples from 15 arcsec/min rotation
Detector nonlinearity	Continuous	Well below saturation; calibration source monitors gain

All auxiliary data (temperature, beam intensity, stage position, time stamps, detector images) recorded for post-hoc correction. Raw data and environmental parameters will be made publicly available upon publication.

6. Null Hypothesis, Kill Condition, and Optional MSDS Interpretation

SECTION 6 CONTAINS OPTIONAL MSDS INTERPRETATION. The null hypothesis test (below) is model-independent. The MSDS-specific interpretation of a positive or negative result is clearly labeled. Acceptance or rejection of the MSDS framework does not affect the experimental validity.

6.1 Null Hypothesis

The null hypothesis is that I(t) is constant aside from random noise — consistent with spatial rotational invariance. The data are fitted to I(t) = I_0 + DI cos(omega_sid t + phi) with DI as a free parameter. Non-detection rules out DI/I_0 > 10^-5 at 95% confidence.

6.2 Kill Condition

KILL CONDITION (as specified in directive): 'A null result down to 10^-5 relative intensity provides a definitive Kill Condition for local electron-sector coupling at the v^2 scale, necessitating a re-evaluation of the MSDS interaction strength.' Specifically: non-detection at 10^-5 constrains the torsion offset in the electron sector to be less than 2×10^-5 — an order of magnitude below the cosmological value inferred from galaxy clustering (Series E, doi:10.5281/zenodo.1082196).

6.3 Optional MSDS Interpretation

Within the MSDS vacuum-lattice framework (Series G, doi:10.5281/zenodo.19456798), the predicted modulation amplitude at the v^2 scale is DI/I_0 ~ $v^2/2$ = 85 ppm = 8.5×10^-5 — an order of magnitude above the experiment's target sensitivity. A positive detection would provide a directional anisotropy amplitude mappable to the SME c_μν electron-sector coefficients (Appendix B). A null result at 10^-5 would constrain the MSDS torsion-electron coupling independently of cosmological observations.

SERIES H — EXPERIMENTAL DESIGN SUMMARY

Observable: sidereal modulation of forbidden (222) reflection in nanodiamonds

Method: RDAC (rotating signal cell) + stationary reference cell (bulk diamond)

Key innovation: 1 Hz active polarization flipper + lock-in at omega_sid

Sidereal/solar discrimination: 3.9-min phase shift per cycle

Sensitivity: $DI/I_0 \sim 10^{-5}$ | Conservative SNR~2 | Optimistic SNR~21

Null result (10^{-5}): rules out $DI > 10^{-5}$ and constrains MSDS v^2 electron coupling

Positive result: first sidereal anisotropy in crystallography; SME $c_{\mu\nu}$ mapping

Beam time: 11 shifts / 88h | Suitable beamlines: APS 13-ID, ESRF ID15, SPring-8 BL10XU

The null hypothesis is that the sky is isotropic. This experiment tests it with a diamond.

Appendix A: Detailed Sensitivity Calculation

Assumptions (APS 13-ID / ESRF ID15): incident flux 10^{12} photons/s in 100 µm spot; (222) count rate 1e+05 photons/s; integration time 60 s.

$\sigma_{shot}/I = 1/\sqrt{1e+05 \times 60} = 4.08e-04$

Lock-in gain over 72h (1h bins): $\sqrt{72} = 8.5$ [1-min bins give $\sqrt{4320} = 65.7$]

Conservative: $\sigma_{final} = 4.0e-05/8.5 = 4.71e-06 \rightarrow SNR(1e-05) = 2.1$

Optimistic: $\sigma_{final} = 4.0e-06/8.5 = 4.71e-07 \rightarrow SNR(1e-05) = 21.2$

Appendix B: SME Mapping (Optional — if positive signal found)

If a sidereal modulation is detected, its amplitude can be mapped to SME electron-sector anisotropy coefficients $c_{\mu\nu}$. The expected bound is $|c_{xx} - c_{yy}| \sim 10^{-15}$ GeV, comparable to current atomic clock limits but in a completely untested crystallographic sector. A detailed mapping is deferred to a follow-up paper if a positive signal is found. Reference: Colladay & Kostelecky, Phys.Rev.D 58, 116002 (1998).

References

[1] Kostelecky V.A., Russell N. Rev.Mod.Phys. 83, 11 (2011). SME data tables.

[2] Herrmann S. et al. Phys.Rev.D 80, 105011 (2009). Lorentz invariance test.

[3] Colladay D., Kostelecky V.A. Phys.Rev.D 58, 116002 (1998). SME framework.

[4] Zheng J. et al. PNAS 120, e2308340120 (2023). Nanodiamond (222) reflection.
[5] Su J. et al. ACS Appl.Nano Mater. 5, 15678 (2022). Nanodiamond structure.
[6] Hirai H. et al. Phys.Rev.B 43, 5457 (1991). High-pressure carbon phases.
[7] Mochalin V.N. et al. Nat.Nanotechnol. 7, 11 (2012). Detonation nanodiamond.
[8] Mao H.K. et al. Rev.Sci.Instrum. 89, 063905 (2018). DAC radial diffraction.
[9] Munday J.N. et al. Phys.Rev.A 80, 023825 (2009). Casimir force measurement.
[10] Aharonov Y. Phys.Rev.Lett. 94, 150401 (2005). Quantum measurement.
[11] Kadunic D. Series G: doi:10.5281/zenodo.19456798. Phenomenological framework.
[12] Kadunic D. Series E v2.0: doi:10.5281/zenodo.1082196. JWST clustering.
[13] Kadunic D. Series D: doi:10.5281/zenodo.1082195. 18.52 nHz resonance.
[14] Kadunic D. Series A: doi:10.5281/zenodo.1082194. QQG core.